

Games Features Guide:

- ❖ **Mods:** When in main menu players will pick mods from cloud to download and only will save the [Names] in **main storage** to be used later when creating a game. When the player creates a new game file, they will choose out of the ones downloaded by grabbing the saved names to be picked. This will save the unity packages in **game storage** for when the player loads the game, it will add all the scripts/stuff in a mod folder to be handled by the game. Mods should be grabbed by our scripts.

- **Tasks:** Load Cloud mods, save properly in unity system, add to game scene every time loaded. Download UI, Pick UI

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- ❖ **Cloud:** Storage Bucket that hold unity packages (mods) along with data regarding them on the cloud. System will be needed in unity to upload mods to the cloud (UI) as well to grab it. The grabbing section will be used by mark in modding section. save items: Unity Package, Name (Key), Creator, Description, Photo (Less important),

- **Tasks:** Upload, Download, UI, Cloud Storage

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- ❖ ~~**Save System:** Save system with 2 storage modes. Main storage that handles global settings like active level, and game storage where it stores game specific data like mods and data:~~

- ~~**Tasks:** MainSave, MainLoad, GameSave, GameLoad~~

MainSave Names	GameSave Names
ActiveLevel	
LevelList	

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- ❖ **Settings:**

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- ❖ **Game Scenes:** Travel Scene, and Game Scene. Travel Scene is on top where travel takes place where player pick points to travel as well as goal destination. Game Scene is where player is shown with player inputs as well as combat, looting, and encounters; It's where the player needs to survive.

- **Tasks:** UI

❖ **Travel:** Top section of game. Players picks a location on map, and the game will then move the player towards it. Includes 2 modes: state and city travel.

- **Target:** Player selects a point on the map where player would like to go. This is given to nav mesh system to generate a path.
- **NavMesh:** Creates a path to target taking optimal path. Can be turned off to create a straight path towards target. If it cannot make a path, then target will not be placed.
- **NavMesh Bool**
- **Path:** List of points to be used by movement system to move player to ultimately reach target. Target is last point.
- **Travel State:** State player is in regarding if the player wants to move.
 - **Moving:** Player is following path towards target.
 - **Paused:** Player stops moving, don't consume food.
 - **Encountered:** When player is in an encounter, prevents player from 'moving'.
 - **Escape:** Only option to leave encounter without completing it, move in opposite direction of path to leave. Encounter will still be there.
- **Player Marker:** visual representation of where player is on map
- **Movement:** if in Moving state and there is a path, move towards first point. If point is reached, remove point and go to next.
- **Encounter Check:** When in moving state, every 5 seconds, there is a 1/12 chance of spawning an encounter. Calls encounter pool to spawn encounter in game scene. Won't spawn if within encounter zone range (no overlapping Encounter zones).
- **Encounter Zones:** When an encounter spawns an encounter zone will be spawned. Represents where an active encounter is. Only will disappear when encounter is complete after a min to make sure encounter don't immediately happen after completing. If re-entered will resume encounter.
- **Hazards**

❖ **State Travel:** subset of travel. Represents country maps.

- **Country Map:** UI map of the USA that is zoomable and zigzag-able (less important)
- **Pathways (less important):** Paths from cities to cities that are faster to travel. Nav-mesh can take paths for optimal routes. Create paths to the 3 closest cities.
- **US States:** Object that will handle code at start like creating cities. Are the US states them selves.
- **Goals:** Target cities to win the game, escape in on off 3 end locations
 - Texas, Florida, or Maine

- **Zones:** Gives different encounters depending on where the player is by giving set of encounters to the encounter pool. Needs to show them visually to the player.
 - **Cities:** Locations to go to gain loot. Spawns list of cities at start of game in each state with rules of how many & distance apart. When player enters a city, then it will switch to City travel mode and add the scene above on the map.
- ❖ **City Travel:** If in a city, can travel on a city map to loot building. Part of general travel system.
- **Buildings:** Locations players may travel to for loot. Assigned a building type that give different loot pools. Won't be shown to the player unless player learn information from possible encounters.
 - **Save Data**
 - **Building Type** {Health, Weapon, Generic, Food}
 - **Known:** show icon if have knowledge knowledge
 - **Looted**
 - **Id**
 - **Roads:** Players may only travel on roads to destination
 - **Exit Locations:** Players will spawn at a random exit location where they must be on to go back to state travel.
 - **City Maps**
- ❖ **Encounters:** when travelling can spawn one in game scene where players need to either handle or escape.
- **Win Condition:** Player completes the encounter and encounter zone is removed from the map.
 - **Escape:** Player choses to leave and encounter information is saved.
 - **Spawning:** Picks an encounter from encounter pool, and spawns' relevant things in game scene.
 - **Re-entry:** Respawns saved info from escape.
 - **Encounter Pool:** multiple lists of multiple encounters to form 1 list to grab an encounter. Lists that will be formed into 1 are based on player position.
 - **Generic Pool:** List that will always be included when grabbing an encounter
 - **State Pool:** List that will be included if player is in state travel
 - **City Pool:** List that will be included if player is in city travel
 - **Zone Pools:** List that will be included for the zone the player is in

- **State & City Zone pools:** Less important

❖ **Looting:** System that can add items to inventory

- **Looting Entities:** Entities can be looted for small resources
- **Looting Obj:** Encounters can have lootable objects (same controls for above)
- **Looting Buildings:** In looting mode when player enters a building, a building scene is generated to include lootable objects, and possible entities.

• Types of Loot:

- **Weapons:** Guns, melee weapons, ammo, and attachments.
- **Supplies:** Water, food, health items, scrap (general stuff), money.
- **Player Upgrades:** Items that improve player stats like health or strength.

❖ **Combat:** Player will use keyboard WASD to move the player and use the mouse to shoot. Additional keyboard inputs to help the player. Enemies will be attacking the player where the player needs to fight back or run. Needed input for:

- Move
- Shoot
- Reload
- Holster
- Jump
- Swap weapon
- Players can equip 2-gun slots that can be swapped between
- Weapon dmg and other values can be modified by other stats (attachments)

❖ **Weapon combat:** Items that can be used for combat. Can be found while looting.

- **Guns:** Need system to fire them as well. And be used by AI humans
 - **Bullet Damage:** Amount of damage a bullet deals upon impact.
 - **Rate of Fire:** Time between each bullet fired. Won't affect semi-autos.
 - **Accuracy:** Where bullets can fire,
 - **Damage Range:** The Distance at which the weapon's bullet damage will drop off (e.g., half damage at a certain distance).
 - **Reload Speed:** Time it takes to reload.
 - **Wait Time:** Number of seconds it takes before the next burst of bullets can be fired. This cooldown period affects burst and semi auto weapons.
 - **Burst:** Number of bullets fired in a single burst before the player must wait (related to the Wait Time).

- **Clip Size:** Maximum number of bullets the weapon can hold in a single magazine before a reload is needed.
- **Ammo Type:** Type of ammunition used by the weapon. Related to resource management.
- **Semi-Auto:** Boolean value determining if the weapon is semi-automatic. If true, the player must click to fire each bullet.
- **Weight:** Affects inventory holding capabilities.
- **Fired Bullets:** Number of bullets fired at once (for shotguns).
- **Penetration:** How many entities bullets can go through.
- **Attachment Slots:** List of attachment slots.
- **Type:** weapon type for simplicity
- **Melee**
 - **Damage (Dmg):** The amount of damage the melee weapon inflicts with each strike.
 - **Attack Speed:** How quickly the player can swing the melee weapon.
 - **Weight:** Affects inventory holding capabilities.
 - **Durability:** How many hits till break
 - **Attachment slots:**
- **Attachments:** Tools that can be added to guns. Can be found while looting. Affects weapon stats based on attachment.

Attachment OBJ:

- Weight
- Type
- Affected stats.

Types:

- Barrel
- Scope
- Mag
- Grip
- Blunt
- Blade
- Handle

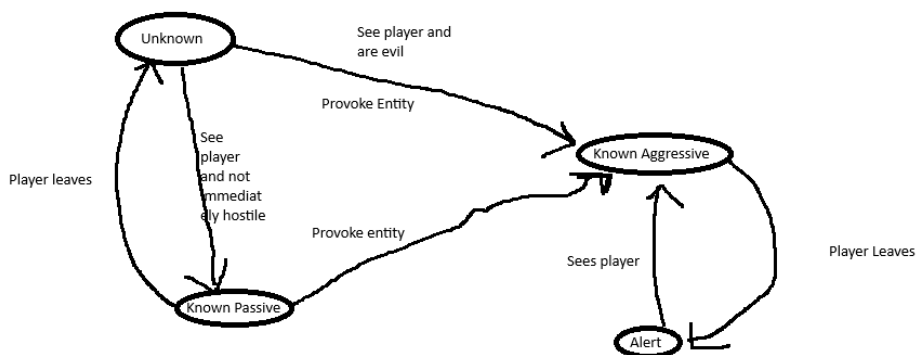
❖ **Enemies:** Entities that can attack the player during combat. **Awareness**

- **Zombies:** Will attack the player within close range.
 - **AI:** Closest human that is seen is target, move towards target and attack target. If not become unaware.
 - Health, Speed, Damage
 - Awareness Level (do they know player location)

Core part of enemy encounters and combat. They are designed with three key stats: Health, Speed, and Damage. Their AI is intentionally simple: braindead and aggressive; focused solely on charging directly at the player without any strategic thinking. They use Navigation Meshes to assist with movement and ensure they can reach the player effectively, even around obstacles. Zombie Types Include:

	Health	Speed	Damage
Zombie	Normal	Normal	Normal
Runner	Low	High	Normal
Heavy	High	Low	High
Tank	Very High	Low	Normal
Bleeder (less important)	Normal	Low	Very High
Weaker (less important)	Low	Normal	Low

- **Humans:** Will attack the player over close or long range.
 - **AI:**
 - Health, Speed
 - Awareness Level (do they know player location)
 - **Unknown passive:** Don't know where player is, do normal ai
 - **Unknown aggressive**
 - **Known Passive:** Know where player is but wont attack
 - **Known Aggressive:** Know where player is and will attack
 - **Alert:** Don't know where player is but when seen will attack

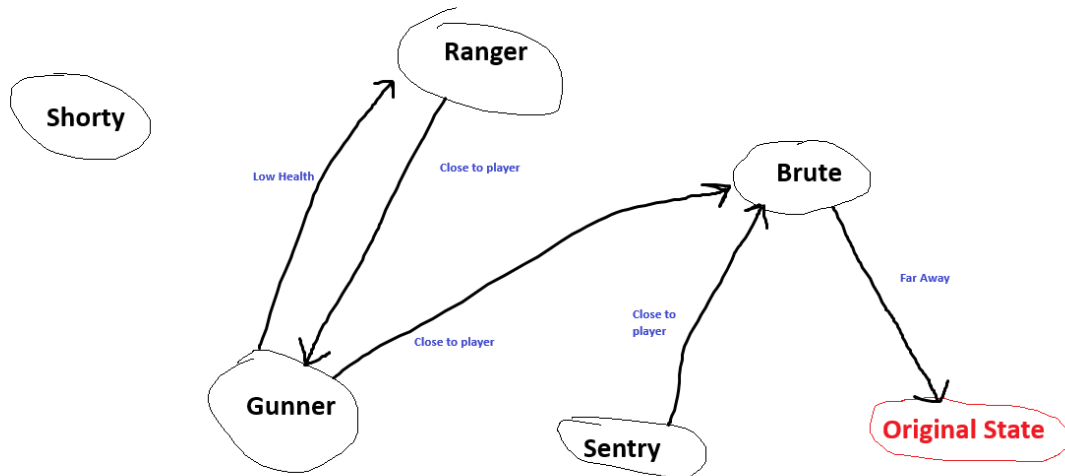


Enemies play a key role in combat encounters, actively attempting to harm the player. Their behavior is governed by finite state machines, allowing for dynamic decision-making based on their type, distance to the player, and current health. Each enemy starts with their own state but allows change. Will keep track of original state to go back to. Enemy Types:

- **Shorty** – Prefers close-range combat, getting near the player to fire. Typically equipped with shotguns.
- **Ranger** – Fights from a long distance, keeping away from the player while shooting.

- **Gunner** – The default enemy type. Maintains a moderate distance while engaging the player.
- **Sentry** – Stationary enemies that remain in one spot, firing from their fixed position.
- **Brute** – Melee-focused enemy that charges directly into close-quarters combat.

Each enemy type adjusts its behavior dynamically switching states based on factors like **health** and **proximity to the player**. Navigation Meshes will support their movement, allowing enemies to navigate terrain intelligently. Swapping to brute equips melee and re swaps when going back to original state.



- ❖ **Survivors:** Humans that can be seen during Encounters that can join the party
 - Survivors are NPCs that will accompany the player, helping in combat and providing additional support. Survivors have their own health and can die during encounters.
 - If the main character dies, control can shift to a survivor by choice, ensuring continuity in gameplay. Stats are set back to default if switched. (Less important)
 - Each party member will take a bit of food when player consumes. Health will be given when player uses healing item (Heals both). They have a lot more health than the player to make sure they don't die easily and only have a gun inventory slot with unlimited ammo.

AI Behavior: Stay close to player, if an enemy is within range shoot at closest. Includes Navigation Meshes to stay close to the player when a bit far and if outside certain range teleport to player to ensure not getting lost.

Good, bad, neutral. Spawned during encounters.

- ❖ **Game Scene:**

The Game scene is on the bottom section of the game and is where the gameplay is located. This will include the player walking and encounters. Also, when the player opens the player menu, it will open a page over it.

❖ **Player Movement/Buttons:**

- Movement: **WASD**
- Looting: **E**
- Shooting: **Left Click**
- Switch Weapons: **Q**
- Reload: **R**
- **Holster/Safety: H** (don't accidentally shoot during encounter)

❖ **Inventory System:** Weight based system with different tab sections in player menu.

- **Player Stats Tab:** For player stats, not inventory
 - **Health:** Level that affects health. Default is 100 at level 1 where each level gives **10 hp**
 - **Speed:** level at which the player moves, Level 1 is speed 1 where each level gives **0.05** which will get multiplied in travel speed and game walking speed.
 - **Strength:** How much player can hold, and melee damage can be done. Default 1 allows the player to hold **50kg** in inventory where each level gives **5kg**. Melee damage starts at 1 with each level giving **0.05** which will be multiplied by melee damage.
 - **Dexterity:** Will affect reload speed and melee speed. Lv. 1 starts at 1 with each level adding **0.05** which will be multiplied into reload and melee speed.
 - **Charisma/Luck:** Will affect chances of encounters and finding more loot
 - **Level:** Each time a player levels up, may add one level to a field. Player starts game at lv. 1 with **3** point to spend.
- **Combat tab**
 - **2 equipped weapons**
 - **Armor**
 - **Weapon Section**
 - Guns
 - Melee
 - **Attachments Section**
 - **Ammo Section**
- **Health & Consumables**
 - **Health Section**
 - **Consumable Section**
 - Water

- Food
- Coffee
- Items Tab & Other Tab
 - Utilities
 - Other

The inventory is limited by weight. Each item in the game has a weight value, and players can only carry as much as their strength statistics allow. Players also have two weapon slots that don't affect the weight but hold weapons that can be easily accessed in combat. Items will have a category associated with them to allow finding.

❖ Items:

Type {Gun, Melee, Attachment, Ammo, Med, Water, Food, Coffee, Utility, Other} (for tab section)

- Guns and melee can be equipped and used for fighting.
- Attachments can be dragged on to weapons
- Ammo will be used automatically when firing bullets
- Medicine can be used when clicked on to give health
- Water and Food will be eaten automatically when hungry/thirsty
- Coffee can be drunken when clicked on to give speed boost + give tired meter a small boost
- Utilities can be used automatically depending on what it does or be clicked on
- Other: to be determined

Lists:

❖ Encounters

❖ Items

Weapons, Boat, Coffee, water, food, Small health kit, Medium health kit, big health kit,

❖ Weapons

G u n s	B u l l e t D m g	R a t e o f f i r e	Acc ura cy	D m g r a n g e	Re lo ad Sp ee d	W ai t t i m e	B u r s t	C l i p S i z e	A m m o T y p e	S e m i - A u t o	W e i g h t	V i s i o n	F i r e d B u l l e t s	P e n e t r a t i o n	Att ac h m e n t s	T y p e

Melee	Damage	Attack Speed	Type	Weight	Durability	Attachment Slots

❖ Attachments

Item names:

Full auto riffles	AK46	AR14	Swat-AR
Semi Auto Riffles	AR12	Safety-Riffle	
Burst riffles	M15	Service-AR	
Shotguns	Pump Shorty	Auto Shorty	Double barrel

Pistols	M1911	Revolver	PK27
SMGS	MP13	MP5L	G10
Snipers	BoltAction	HeavySniper	
Melee	Knife	Sword	pipe
Armor	LightArmor	MediumArmor	HeavyArmor
Meds	LightMed	MediumMed	HeavyMed
Food	LightSnack	Snack pack	FullMeal
Other	SpeedBoost	XPDrink	

Ak46, AR14,

M16,